

JACKSON MORRIS

CURRICULUM VITAE

Department of Mathematics
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EMPLOYMENT

Postdoctoral Researcher
Vanderbilt University

January 2027-August 2029

Postdoc SLMath
SLMath Semester Program
Motivic Homotopy Theory: Connections and Applications

August-December 2026

EDUCATION

Ph.D. in Mathematics
University of Washington
Advisors: Kyle Ormsby and John Palmieri
Thesis title: Cooperations in motivic homotopy theory

September 2021-June 2026

M.S. in Mathematics
University of Washington

September 2021-June 2023

B.S. in Mathematics, magna cum laude
University of Kentucky

September 2017-December 2020

RESEARCH INTERESTS

Motivic homotopy theory, chromatic homotopy theory, hermitian K-theory, algebraic K-theory, synthetic spectra, nilpotence and periodicity, Adams spectral sequence, periodic self-maps, redshift and blueshift

PUBLICATIONS AND PREPRINTS

9. *Diachromatic blueshift*, joint with Kyle Ormsby. In preparation.
8. *Motivic K-theory cooperations over the rationals*, joint with Sarah Petersen and Liz Tatum. In preparation.
7. *An exotic self-map of periodicity 1*. In preparation.
6. *Periodic phenomena in stable motivic homotopy theory*. In preparation.
5. *Immersion of C_2 -projective spaces via K $\mathbb{R}$ -theory*, joint with Manyi Guo, Alex Waugh, and Albert Jinghui Yang. Available at <https://arxiv.org/abs/2604.25260>.
4. *Spectrum level splittings of truncated motivic Brown–Peterson cooperations algebras*, joint with Sarah Petersen and Elizabeth Tatum. Submitted. Available at <https://arxiv.org/abs/2509.19542>.
3. *Rings of cooperations for hermitian K-theory over finite fields*. To appear in **Algebraic and Geometric Topology**. Available at <https://arxiv.org/abs/2509.02786>.
2. *On the ring of cooperations for real hermitian K-theory*. To appear in **Annals of K-theory**. Available at <https://arxiv.org/abs/2506.16672>.
1. *Toric Double Determinantal Varieties*, joint with Alexander Blose, Patricia Klein, and Owen McGrath. **Communications in Algebra** 49 (2020), 7, 3085–3093.

SELECTED PRESENTATIONS & SEMINAR TALKS

Conference and Colloquium Talks

“Higher Witt K-theories”

AMS Sectional: Computational Homotopy Theory (Tempe, AZ)
Young Topologists Meeting 2026 (Copenhagen, DK)
Cascade Topology Seminar (Portland, OR)

November 2026

July 2026

May 2026

“Splittings and cooperations in motivic homotopy theory”	
Joint Math Meetings 2026 (Washington, D.C.)	January 2026
“On the ring of cooperations for real hermitian K-theory”	
European Talbot 2025 (Kolding, DK)	July 2025
Graduate Student Geometry & Topology Conference 2025 (Bloomington, IN)	April 2025
Joint Math Meetings 2025 (Seattle, WA)	January 2025
European Autumn School in Topology (Utrecht, NL)	September 2024
Motivic Homotopy, K-theory, and Modular Representations (Los Angeles, CA)	August 2024
“Finding Fixed Points with Invariants”	
Reed College Mathematics Colloquium	February 2025
<u>Seminar Talks</u>	
University of Washington Topology Seminar	February 2026
University of Illinois Urbana-Champaign Topology Seminar	January 2026
University of Virginia Topology Seminar	November 2025
University of Kentucky Topology Seminar	November 2025
Duke University Geometry and Topology Seminar	October 2025
University of Colorado Boulder Homotopy Theory Seminar	September 2025
University of Michigan Geometry Seminar	September 2025
University of Notre Dame Topology Seminar	September 2025
University of Washington Topology Seminar	June 2025
<u>Contributed Talks</u>	
eCHT Reading Seminars	
“An upper bound on the degree of symmetry of an exotic sphere”	March 2026
“Higher Adams differentials and hidden extensions”	November 2025
“Motivic twisted K-theory”	April 2024
Graduate Student Seminars	
“Higher real K-theories, redshift, and blueshift”	May 2026
“Filtered objects, spectral sequences, and t -structures”	January 2026
“(A synthetic approach to detecting) v_1 -periodic families”	November 2025
“Algebraic and Milnor K-theory”	October 2025
“Galois descent and the Picard group of K-theory”	May 2025
“The Adams spectral sequence and Hopf algebroids”	February 2025
“Symplectic orientations”	August 2024
Workshops	
“ $\pi_* \mathrm{THH}(\mathbb{Z}_p^{\mathrm{fil}})/p$ and $\pi_* \mathrm{THH}(\ell_p^{\mathrm{fil}})/(p, v_1)$ ” (Talbot, Nacogdoches TX)	August 2024
“Hochschild Homology” (European Autumn School in Topology, Utrecht, NL)	September 2024

SERVICE AND PROFESSIONAL ACTIVITIES

Refereeing and Reviewing

Algebraic and Geometric Topology	2025-Present
ZBMath	2026-Present

Seminar Organization

DUBTOP Seminar	2025-2026
Topics included: Chromatic homotopy theory, stacks, Goodwillie calculus	
Back-To-School seminar	Fall 2024
Student lectures on fun math learned over the summer.	
Reading seminars	2023-2024
Topics included: Algebraic K-theory, cohomology operations, motivic homotopy theory, vector bundles and topological K-theory, categorical homotopy theory, model categories	

TEACHING EXPERIENCE

Instructor of Record

University of Washington

Math 480 – Homotopy theory

Math 208 – Linear algebra

Spring 2026

*Winter 2023, Summer
2023*

Graduate Assistant

Electronic Computational Homotopy Theory (eCHT)

Research Seminar

2025-2026

Quadratic Curve Counting

Fall 2024

Park City Math Institute (PCMI)

Summer 2024

Experimental Math Lab

University of Washington

Fall 2021 - Spring 2026

Courses included: Precalculus, Differential calculus, Multivariable calculus, Linear algebra, Introduction to proofs, Topology

Directed Reading Mentor

University of Washington

2022-2025

Topics included: Homotopy theory, Representation theory, Geometric group theory, Homological algebra

Math Olympiad Judge

UW Math Hour

2023 - 2024

Judge for elementary school math competition

HONORS AND AWARDS

Lee and Cozette McFarlan Travel Fellowship

Summer 2025

University of Washington

\$750

For travel to European Talbot in Kolding, DK.

Research Assistantship

Summer 2025

University of Washington

For good research progress.

JC Eaves Scholarship

2020

University of Kentucky

\$2000

For continued excellence in mathematics as a Kentuckian.

Carolyn Bunyan Award

2020

University of Kentucky

\$2000

For mathematics majors interested in continuing education in mathematics.

SELECTED WORKSHOP AND CONFERENCE PARTICIPATION

Extended Research Visits

SLMath Semester Program: Motivic Homotopy Theory (Berkeley, CA)

Fall 2026

PCMI: Motivic Homotopy Theory (Park City, UT)

Summer 2024

Conference Participation

AMS Sectional: Computational Homotopy Theory (Tempe, AZ)

November 2026

Algebraic Structures in Topology (San Juan, PR)

July 2026

Young Topologists Meeting (Copenhagen, DK)

June/July 2026

Cascade Topology Seminar (Portland, OR)

May 2026

(∞, ∞) -Categories Workshop (Chicago, IL)

March 2026

JMM 2026 (Washington, D.C.)

January 2026

Midwest Topology Seminar (Minneapolis, MN)

October 2025

European Talbot (Kolding, DK)

July 2025

WAGS Spring 2025 (Vancouver, CAN)

April 2025

GSTGC 2025 (Bloomington, IN)	<i>April 2025</i>
JMM 2025 (Seattle, WA)	<i>January 2025</i>
SLMath Hot Topics: Life after the Telescope Conjecture (Berkeley, CA)	<i>December 2024</i>
European Autumn School in Topology (Utrecht, NL)	<i>September 2024</i>
Talbot 2024 (Nacogdoches, TX)	<i>August 2024</i>
Motivic Homotopy, K-theory, and Modular Representations (Los Angeles, CA)	<i>August 2024</i>
eCHT Research Workshop on Hopf Rings (Online)	<i>June 2024</i>
JMM 2024 (San Francisco, CA)	<i>January 2024</i>
Scissors Congruence, Algebraic K-theory, & Trace Methods (Bloomington, IN)	<i>July 2023</i>
Cascade Topology Seminar (Vancouver, CAN)	<i>April 2023</i>
PRIMA Congress (Vancouver, CAN)	<i>December 2022</i>
Graduate Students in Geometry, Topology, and Algebra (Philadelphia, PA)	<i>May 2022</i>

REFERENCES

Mark Behrens John and Margaret McAndrews Professor of Mathematics University of Notre Dame <i>Relationship: Research mentor</i>	mbehren1@nd.edu
Kyle Ormsby Mathematics Department Chair Reed College <i>Relationship: PhD advisor and research collaborator</i>	ormsbyk@reed.edu
John Palmieri Professor of Mathematics University of Washington <i>Relationship: PhD advisor; Mathematics Department Chair during first 2 years of graduate study</i>	jpalmier@uw.edu
J.D. Quigley Assistant Professor of Mathematics University of Virginia <i>Relationship: Research mentor; Program supervisor and mentor at PCMI</i>	mbp6pj@virginia.edu
Charles Camacho Assistant Teaching Professor University of Washington <i>Relationship: Teaching supervisor and mentor</i>	camachoc@uw.edu
Natalie Naehrig Assistant Teaching Professor University of Washington <i>Relationship: Teaching supervisor and mentor</i>	naehrn@uw.edu